

Software Requirements Specification

for

Job Board Platform

Document title	Software Requirements Specification – Job Board Platform
Version	1.0
Prepared by	Muhammad Asad
Course / Lab	Software Construction – Lab #3 (Software Requirements Specification Document)
Institution	Bahria University, Karachi Campus
Date	21/09/2026
Status	Draft for instructor review

Revision history

Version	Date	Author	Description
1.0	21/09/2026	Muhammad Asad	First complete release of the SRS.

Contents

1. Introduction
2. Overall Description
3. Specific Requirements
4. System Features
5. External Interface Requirements
6. Non-Functional Requirements
7. Data Requirements and Use Cases
8. Constraints
9. Assumptions and Dependencies
10. Appendices and Glossary

1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) of the Job Board Platform. It states what the system shall do and how well it shall do it, so that the project developer, the mentor, the instructor and the testers share one understanding of the product before and during construction.

The document is written for four readers: developers, who use it to build and change the software; testers, who derive test cases from it; the instructor and mentor, who review it for completeness; and future trainees, who can use it as a worked example.

1.2 Scope

The Job Board Platform is a web application that connects employers with job seekers. Companies publish job openings, job seekers search for jobs and apply with a cover letter and a PDF resume, and administrators verify companies and control the data of the platform.

In scope for version 1.0:

- Role-based registration, login, Google Sign-In for Job Seekers and password reset by email.
- Company creation, company joining by registration number and administrator verification of companies.
- Job posting, editing, closing, reopening and deletion by verified companies.
- Job search, filtering, job details and bookmarks for Job Seekers.
- Job applications with resume upload, application tracking and applicant review.
- A skill catalogue that links Job Seekers and jobs.
- An administrator dashboard and a Payload CMS admin panel that work on the same data.
- Validation, error handling and request logging.

Out of scope for version 1.0: online payment for job postings, in-app chat between users, video interviews, resume builders, native mobile applications and import of jobs from other job sites.

1.3 System Overview

The system has three tiers. A React single-page application (frontend) runs in the browser and communicates over HTTPS with a Django REST Framework application (backend). The backend stores data in a PostgreSQL database and writes request logs to a second PostgreSQL database. A Payload CMS application gives administrators a second interface that reads and writes the same data through the backend API.

Component	Description
Frontend	React with Vite, Tailwind CSS, React Router, Axios and TanStack Query. Runs in the user's browser.

Backend API	Django and Django REST Framework with JWT authentication. Five modules: accounts, companies, jobs, job applications and skills.
Main database	PostgreSQL. Stores users, companies, jobs, applications and skills.
Log database	A separate PostgreSQL database that stores one record for every API request.
CMS admin panel	Payload CMS. Gives administrators job board views that forward every action to the backend API.

1.4 Document Conventions

- The word shall marks a mandatory requirement. The word should marks a recommendation.
- Every requirement has a unique identifier: FR (functional), NFR (non-functional), BR (business rule), UC (use case) and TC (test case).
- Priority values are Critical (system cannot work without it), High, Medium and Low.
- Each requirement statement contains one requirement and a measurable or checkable condition.

1.5 References

- ISO/IEC/IEEE 29148:2018, Systems and software engineering – Requirements engineering.
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Software Construction Lab 03 manual: Software Requirements Specification Document.
- Django REST Framework, SimpleJWT and Payload CMS documentation.
- Project source repository: github.com/asad594/bytecorp-training-tasks.

2. Overall Description

2.1 Product Perspective

The Job Board Platform is a new, self-contained product. It depends on three external services: Google Identity Services for Google Sign-In, an e-mail delivery service for password-reset messages, and cloud hosting for the frontend and backend. Section 5 describes each interface.

2.2 Product Functions

- Create accounts and log in as Job Seeker, Company Representative or Administrator.
 - Create companies, join companies and verify companies.
 - Publish, edit, close, reopen and delete jobs.
 - Search, filter, view and bookmark jobs.
 - Apply to jobs, track applications and review applicants.
 - Maintain a catalogue of skills and link skills to profiles and jobs.
 - Administer users, companies, jobs and skills through the dashboard and the CMS.
 - Record every API request for audit and troubleshooting.
-

2.3 User Classes and Access

User	Main responsibility	Permitted access
Visitor	Views the public landing page and opens the registration or login pages.	Landing page, registration pages, login pages, password-reset pages.
Job Seeker	Searches for jobs, applies to jobs and follows the status of applications.	Job list and details, own applications, own profile, skill catalogue (read only).
Company Representative	Represents one company, posts jobs and reviews applicants.	Company dashboard, jobs and applications of own company, skill catalogue (read only).
Administrator	Verifies companies and controls users, jobs and skills.	Admin dashboard and CMS admin panel with access to all data and to the logs.

All four classes use a standard web browser and need no technical training beyond basic computer skills. Administrators know the verification procedure of the organization.

2.4 Operating Environment

- Client: a computer, tablet or phone with a current web browser (see NFR for compatible browsers) and an internet connection.
- Server: Python 3.13 with Django 6.0 and Django REST Framework, PostgreSQL, running on a cloud host; frontend served as static files from a content delivery network.
- CMS: Node.js runtime for Payload CMS with its own small PostgreSQL database for CMS administrator login.

2.5 Main Workflow

Step	Name	Description
01	Register	A Company Representative registers, creates a company and receives the status “unverified”. A Job Seeker registers or signs in with Google.
02	Verify	An Administrator reviews the company registration number and verifies the company.
03	Publish	The Company Representative posts a job with its details and required skills.
04	Apply	The Job Seeker finds the job, submits a cover letter and a PDF resume, and the application receives the status Pending.
05	Review	The Company Representative reads the application and sets the status to Reviewed, Accepted or Rejected. The Job Seeker sees the new status.

3. Specific Requirements

3.1 Functional Requirements

The tables below list the functional requirements grouped by area. The column “Source” names the stakeholder who needs the requirement.

3.1.1 Account and Authentication

ID	Requirement	Priority	Source
FR-001	The system shall allow a visitor to register as a Job Seeker by submitting name, email address and password on the Job Seeker registration page.	Critical	Job Seeker
FR-002	The system shall allow a visitor to register as a Company Representative by submitting name, email address and password on a separate Company Representative registration page.	Critical	Company Rep

FR-003	The system shall provide a separate login page for each role: Job Seeker, Company Representative and Administrator.	Critical	All users
FR-004	The system shall reject a registration when the submitted email address is already registered under any role.	High	System
FR-005	The system shall reject a login attempt when the role of the account differs from the role of the login page used.	High	System
FR-006	The system shall issue an access token valid for 15 minutes and a refresh token valid for 7 days after a successful login.	Critical	System
FR-007	The system shall obtain a new access token automatically, without asking for credentials, when the access token has expired and the refresh token is still valid.	High	All users
FR-008	The system shall allow a Job Seeker to register or log in using Google Sign-In.	Medium	Job Seeker
FR-009	The system shall reject Google Sign-In for an email address that is registered as a Company Representative or an Administrator.	Medium	System
FR-010	The system shall send a password-reset link to the registered email address of a Job Seeker or Company Representative who requests it.	High	All users
FR-011	The system shall display the same confirmation message whether or not the submitted email address is registered.	Medium	System
FR-012	The system shall not send a password-reset link to an Administrator account.	Medium	Admin
FR-013	The system shall allow a user who opens a valid reset link to set a new password that satisfies the password rules.	High	All users
FR-014	The system shall delete the stored access and refresh tokens from the client device when a user logs out.	High	All users
FR-015	The system shall provide at least one Administrator account created by the deployment team, because Administrator self-registration does not exist.	Critical	Admin
FR-016	The system shall allow an Administrator to create another Administrator account using a form on the admin dashboard.	Medium	Admin
FR-017	The system shall redirect an unauthenticated user who opens a protected page to the login page.	High	System
FR-018	The system shall open each protected page only for the roles permitted to use it and shall reject all other roles.	Critical	System

3.1.2 Job Seeker Profile

ID	Requirement	Priority	Source
FR-019	The system shall allow a Job Seeker to view their name, email address and selected skills on a profile page.	Medium	Job Seeker
FR-020	The system shall allow a Job Seeker to edit their name and selected skills.	Medium	Job Seeker
FR-021	The system shall allow a Job Seeker to select skills only from the active skill catalogue.	Medium	Job Seeker

3.1.3 Company Creation and Verification

ID	Requirement	Priority	Source
FR-022	The system shall allow a Company Representative to create a company by entering the company name and a unique registration number.	Critical	Company Rep
FR-023	The system shall allow a Company Representative to join an existing company by entering the registration number of that company.	High	Company Rep

FR-024	The system shall reject a join request when no company has the entered registration number.	High	System
FR-025	The system shall mark every newly created company as unverified.	Critical	Admin
FR-026	The system shall prevent the members of an unverified company from publishing jobs.	Critical	Admin
FR-027	The system shall allow a Company Representative to access only the jobs and applications of their own company.	Critical	Admin
FR-028	The system shall allow an Administrator to view the list of companies that await verification.	High	Admin
FR-029	The system shall allow an Administrator to verify and to unverify a company.	Critical	Admin

3.1.4 Job Posting and Maintenance

ID	Requirement	Priority	Source
FR-030	The system shall allow a member of a verified company to create a job with title, description, location, minimum salary, maximum salary, employment type and required skills.	Critical	Company Rep
FR-031	The system shall accept only Full-time, Part-time or Contract as the employment type of a job.	High	System
FR-032	The system shall reject a job whose minimum salary is greater than its maximum salary.	High	System
FR-033	The system shall allow a member of a company to edit the jobs of that company.	High	Company Rep
FR-034	The system shall allow a member of a company to close and to reopen a job of that company.	High	Company Rep
FR-035	The system shall allow a member of a company to delete a job of that company; the deleted job remains stored as an inactive record.	Medium	Company Rep
FR-036	The system shall display to a Company Representative the jobs of their company together with the number of applicants for each job.	High	Company Rep

3.1.5 Job Search and Discovery

ID	Requirement	Priority	Source
FR-037	The system shall display a public landing page with featured jobs and job categories to every visitor.	Medium	All users
FR-038	The system shall display to a logged-in Job Seeker a list of open jobs showing title, company, location, employment type and salary range.	Critical	Job Seeker
FR-039	The system shall filter the job list by the keyword entered in the search box.	High	Job Seeker
FR-040	The system shall filter the job list by the location entered by the Job Seeker.	High	Job Seeker
FR-041	The system shall filter the job list by employment type using the tabs All, Full-time, Part-time and Contract.	High	Job Seeker
FR-042	The system shall exclude closed and deleted jobs from the job list shown to Job Seekers.	High	System
FR-043	The system shall display the complete details of a selected job, including description and required skills.	High	Job Seeker
FR-044	The system shall allow a Job Seeker to bookmark and to remove the bookmark of a job.	Low	Job Seeker
FR-045	The system shall display a loading state, an error state and an empty-result state on every list page.	Medium	All users

3.1.6 Job Applications

ID	Requirement	Priority	Source
FR-046	The system shall allow a Job Seeker to apply to an open job by submitting a cover letter and a resume.	Critical	Job Seeker
FR-047	The system shall accept only resume files in PDF format with a size of at most 5 MB.	High	System
FR-048	The system shall reject a second application from the same Job Seeker to the same job.	High	System
FR-049	The system shall reject an application to a closed job.	High	System
FR-050	The system shall set the status of every new application to Pending.	High	System
FR-051	The system shall allow a Job Seeker to view a list of their own applications with the current status of each application.	High	Job Seeker
FR-052	The system shall prevent a Job Seeker from viewing the applications of other Job Seekers.	Critical	System
FR-053	The system shall allow a Company Representative to view, for each job of their company, the applicants with name, email address, cover letter and a link to the resume.	Critical	Company Rep
FR-054	The system shall allow a Company Representative to set the status of an application of their company to Pending, Reviewed, Accepted or Rejected.	Critical	Company Rep
FR-055	The system shall allow only the applicant and the Company Representatives of the applied company to open a submitted resume.	High	System

3.1.7 Skill Catalogue

ID	Requirement	Priority	Source
FR-056	The system shall allow an Administrator to create, rename and delete skills.	High	Admin
FR-057	The system shall reject a skill whose name matches an existing skill when letter case is ignored.	Medium	System
FR-058	The system shall keep a deleted skill as an inactive record and shall not offer it for new selection.	Medium	System
FR-059	The system shall allow every logged-in user to view the list of active skills.	Medium	All users

3.1.8 Administration (Dashboard and CMS)

ID	Requirement	Priority	Source
FR-060	The system shall allow an Administrator to view all users with name, email address, role and, for Company Representatives, the company and its verification status.	High	Admin
FR-061	The system shall allow an Administrator to ban and to unban a user; a banned user shall not be able to log in.	High	Admin
FR-062	The system shall allow an Administrator to view, edit, close, reopen and delete any job.	High	Admin
FR-063	The system shall allow an Administrator to edit the details of any company, user and skill.	High	Admin
FR-064	The system shall provide a Payload CMS admin panel in which an Administrator can view, edit, verify, ban, close, reopen and delete jobs, companies, users and skills.	Medium	Admin
FR-065	The system shall apply every change made in the CMS admin panel to the same records that the custom admin dashboard uses.	High	Admin
FR-066	The system shall require Administrator authentication before the CMS admin panel opens any job board data.	Critical	Admin

3.1.9 Validation, Error Handling and Logging

ID	Requirement	Priority	Source
FR-067	The system shall validate every submitted field on the server and return an error message that names each invalid field.	High	System
FR-068	The system shall return the HTTP status codes 400, 401, 403, 404 and 500, each with a JSON error message, for failed requests.	High	System
FR-069	The system shall record every API request in a separate log database with correlation ID, method, path, status code, duration and the request and response bodies and headers.	Medium	Admin
FR-070	The system shall replace password and token values with a redaction marker before it stores a log entry.	High	Admin

3.2 Business Rules

ID	Business rule
BR-01	An email address belongs to exactly one account and exactly one role.
BR-02	Administrator accounts are created only by the deployment team or by another Administrator.
BR-03	Only a verified company may publish jobs.
BR-04	The registration number of a company is unique.
BR-05	A Company Representative may read and change only the jobs and applications of the company to which the representative belongs.
BR-06	A Job Seeker may read only the applications that the Job Seeker submitted.
BR-07	A Job Seeker may apply to the same job only once.
BR-08	The minimum salary of a job is not negative and is not greater than its maximum salary.
BR-09	Jobs and skills are never physically removed; deletion marks the record as inactive so that history and audit data stay complete.
BR-10	A resume is a PDF file of at most 5 MB.

3.3 Measurable Acceptance Criteria

Each row below states how a tester decides whether the selected requirement is met.

Requirement	Acceptance criterion
FR-001	Submitting a valid name, email address and password returns HTTP 201 and the new account can log in immediately. A second registration with the same email returns HTTP 400 with an email error.
FR-006	The decoded access token expires 15 minutes ($\pm$ 5 seconds) after issue and the refresh token expires 7 days after issue.
FR-010	A reset e-mail reaches the inbox of a registered Job Seeker within 2 minutes of the request.
FR-026	A member of an unverified company receives HTTP 403 when creating a job. After verification, the same request returns HTTP 201.
FR-032	A job with minimum salary 90000 and maximum salary 50000 returns HTTP 400 and is not stored.
FR-039	A search for a keyword returns only jobs whose title contains that keyword, and the list refreshes within 1 second for 95% of searches.
FR-047	A 4.9 MB PDF is accepted. A 5.1 MB PDF and a .docx file are both rejected with an error message that names the problem.
FR-054	A representative of company A receives HTTP 403 when changing the status of an application that belongs to company B.
FR-057	Creating the skill "python" when "Python" exists returns HTTP 400.
FR-069	After any API call, one log record with the same correlation ID exists in the log database within 5 seconds.

4. System Features

Each feature groups related requirements. The table of a feature states its purpose, priority, inputs, processing, outputs, exceptions and the linked requirements.

4.1 Role-Based Authentication and Account Access

Purpose	Lets Job Seekers, Company Representatives and Administrators enter the system through role-specific pages, and lets users recover a forgotten password.
Priority	Critical
Inputs	Name, email address, password, Google ID token, password-reset token, new password.
Processing	Validate fields; check that the email is unused; hash the password; compare the account role with the login page role; issue access and refresh tokens; verify the Google ID token; create and e-mail a reset link.
Outputs	Access and refresh tokens, user profile data, confirmation messages.
Exceptions	Duplicate email: registration rejected. Wrong role: login rejected. Expired or invalid reset link: error message. Google email registered under another role: HTTP 403.
Linked requirements	FR-001 to FR-018

4.2 Company Creation and Verification

Purpose	Lets employers represent a company and lets Administrators confirm that a company is genuine before it posts jobs.
Priority	Critical
Inputs	Company name, registration number, join request number, verification decision.
Processing	Check that the registration number is unique; create the company as unverified and link the creator as member; link joining representatives; change the verification flag.
Outputs	Company record, membership record, verification status shown on the dashboard.
Exceptions	Duplicate registration number: creation rejected. Unknown registration number: join rejected. Verification attempt by a non-administrator: HTTP 403.
Linked requirements	FR-022 to FR-029

4.3 Job Posting and Maintenance

Purpose	Lets verified companies publish and maintain job openings.
Priority	Critical
Inputs	Title, description, location, minimum and maximum salary, employment type, required skills, status change.
Processing	Check company membership and verification; validate the fields; store the job with the status Open; update or mark the job inactive on request.
Outputs	Job record, updated job list of the company with applicant counts.
Exceptions	Unverified company: HTTP 403. Salary range invalid: HTTP 400. Job of another company: HTTP 403.
Linked requirements	FR-030 to FR-036

4.4 Job Search and Discovery

Purpose	Lets Job Seekers find suitable open jobs quickly.
Priority	Critical
Inputs	Keyword, location, employment type tab, bookmark action.
Processing	Load open jobs from the API; filter by keyword, location and employment type; show loading and empty states; keep the bookmark list.
Outputs	Filtered job cards and the job detail view.
Exceptions	API not reachable: error state with a retry option. No matching job: empty-result message.
Linked requirements	FR-037 to FR-045

4.5 Job Application and Tracking

Purpose	Lets a Job Seeker apply to a job and follow the status of the application.
Priority	Critical
Inputs	Job, cover letter, PDF resume.
Processing	Check that the job is open and not yet applied to; check file type and size; store the file and the application with the status Pending.
Outputs	Confirmation message and a new entry in My Applications.
Exceptions	File not PDF or larger than 5 MB: rejected. Second application to the same job: rejected. Closed job: rejected.
Linked requirements	FR-046, FR-047, FR-048, FR-049, FR-050, FR-051, FR-052

4.6 Applicant Review

Purpose	Lets a Company Representative read applications and record the hiring decision.
Priority	Critical
Inputs	Selected job, new application status.
Processing	Check company ownership; list applicants with cover letter and resume link; save the new status.
Outputs	Applicant list; updated status visible to the Job Seeker.
Exceptions	Application of another company: HTTP 403. Missing resume file: message shown, status change still possible.
Linked requirements	FR-053, FR-054, FR-055, FR-027

4.7 Skill Catalogue

Purpose	Provides one shared list of skills for Job Seeker profiles and job requirements.
Priority	High
Inputs	Skill name; skills selected by users.
Processing	Compare names ignoring letter case; store skills; mark deleted skills inactive; offer only active skills.
Outputs	Skill list, skill selector on the profile page and on the job form.
Exceptions	Duplicate name: HTTP 400. Non-administrator creates a skill: HTTP 403.
Linked requirements	FR-056 to FR-059, FR-019, FR-020, FR-021

4.8 Administration through Dashboard and CMS

Purpose	Gives Administrators full control of users, companies, jobs and skills in two interfaces that use the same data.
Priority	High
Inputs	Administrator credentials, verification, ban, edit, close, reopen and delete actions.
Processing	Authenticate the Administrator; forward each CMS action to the backend API with an administrator token; apply the change to the shared database.
Outputs	Updated records visible in both the dashboard and the CMS panel.
Exceptions	Non-administrator access: rejected. Backend not reachable: the CMS panel shows an error and changes nothing.
Linked requirements	FR-060 to FR-066

4.9 Validation, Error Handling and Request Logging

Purpose	Keeps stored data correct and gives developers a record of every request.
Priority	Medium
Inputs	Every API request and its data.
Processing	Validate fields; map failures to HTTP status codes; write a sanitized log record to the log database.
Outputs	Field-level error messages; log records with correlation ID.
Exceptions	Log database unreachable: the request still completes and the failure is written to the server log.
Linked requirements	FR-067 to FR-070

5. External Interface Requirements

5.1 User Interfaces

Screen	Role	Purpose
Landing page	Visitor	Shows featured jobs and categories and links to registration and login.
Registration and login pages	Job Seeker, Company Rep, Admin	One page pair for each role; Google Sign-In button for Job Seekers.
Forgot and reset password	Job Seeker, Company Rep	Request a reset link and set a new password.
Jobs list	Job Seeker	Search box, location filter, employment type tabs and job cards.
Job detail and apply form	Job Seeker	Full job description, required skills, cover letter field and resume upload.
My applications	Job Seeker	List of submitted applications with status badges.
My profile	Job Seeker	View and edit name and skills.
Company dashboard	Company Rep	Create or join a company, post and maintain jobs, open the applicants of each job.
Admin dashboard	Administrator	Company verification, users, jobs, skills and the form to add an Administrator.
CMS admin panel	Administrator	Payload CMS views for jobs, companies, users and skills.

All screens share one dark navy theme with reusable Button, Input and Badge components. Every screen that loads data shows loading, error and empty states (see FR-045).

5.2 Hardware Interfaces

The system has no dedicated hardware. Users need a device with a web browser and an internet connection. The servers run on virtual machines of the cloud host.

5.3 Software Interfaces

Interface	Description
Backend REST API	JSON over HTTPS under the prefix /api/v1/. Main endpoints are listed in section 5.4.
PostgreSQL	Main database for business data and a separate log database, accessed through the Django ORM.
Google Identity Services	The browser obtains a Google ID token; the backend verifies it at endpoint /api/v1/accounts/auth/google/.
E-mail service	Password-reset e-mail is sent through an HTTPS-based e-mail API because the hosting platform blocks outbound SMTP ports.
Payload CMS proxy	The CMS forwards every administrator action to the backend API with a service-account access token.

5.4 Communication Interfaces and Main Endpoints

All communication uses HTTPS with JSON bodies. The resume upload uses multipart/form-data. Protected requests carry the header Authorization: Bearer <access token>. Cross-origin requests are accepted only from configured origins.

Method	Endpoint	Purpose	Access
POST	/api/v1/accounts/register/{role}/	Register a Job Seeker or Company Representative.	Visitor
POST	/api/v1/accounts/login/{role}/	Log in and receive access and refresh tokens.	Visitor
POST	/api/v1/accounts/auth/google/	Google Sign-In for Job Seekers.	Visitor
GET	/api/v1/accounts/profile/	Read the profile of the logged-in user.	Logged-in user
GET, POST	/api/v1/jobs/	List open jobs; create a job.	Logged-in user; POST for company members
GET, PUT, PATCH, DELETE	/api/v1/jobs/{id}/	Read, edit, close or delete a job.	Logged-in user; changes for owners

6. Non-Functional Requirements

Non-functional requirements state how well the system shall operate. Each requirement contains a measurable value.

ID	Quality area	Measurable requirement	Priority
NFR-001	Performance	The system shall return the response of a read request within 1 second for 95% of requests under a load of 100 concurrent users.	High
NFR-002	Performance	The system shall display the first results of the job list within 3 seconds for 95% of page loads on a 10 Mbps connection.	High
NFR-003	Performance	The system shall complete the upload of a 5 MB resume within 10 seconds on a 5 Mbps connection.	Medium

NFR-004	Security	The system shall store every password only as a salted hash and never as plain text.	Critical
NFR-005	Security	The system shall transmit all data between browser and server over HTTPS.	Critical
NFR-006	Security	The system shall check the role of the user on every API endpoint and return HTTP 401 for unauthenticated and HTTP 403 for unauthorized requests.	Critical
NFR-007	Security	The system shall require passwords of at least 8 characters.	High
NFR-008	Security	The system shall accept browser requests only from the origins listed in its configuration.	High
NFR-009	Security	The system shall use parameterized database queries and output escaping so that testing finds no SQL injection and no stored cross-site scripting vulnerability.	Critical
NFR-010	Security	The system shall read secret keys, database credentials, e-mail service keys and OAuth settings from environment variables and shall not store them in the source repository.	Critical
NFR-011	Security	The system shall check the type and size of every uploaded file before it stores the file.	High
NFR-012	Availability	The system shall be available at least 99.5% of each calendar month, excluding planned maintenance announced 24 hours in advance.	High
NFR-013	Reliability	The system shall show the user a readable error message and shall not display stack traces or internal error details in any response.	High
NFR-014	Usability	The system shall allow a new Job Seeker to complete registration within 2 minutes.	Medium
NFR-015	Usability	The system shall allow a Job Seeker to apply to a job in at most 4 clicks from the job list.	Medium
NFR-016	Usability	The system shall display all pages without horizontal scrolling on screen widths from 360 px to 1920 px.	Medium
NFR-017	Usability	The system shall use the same colour theme, fonts and reusable button and input components on all pages.	Low
NFR-018	Auditability	The system shall record the time, user role, endpoint and outcome of every API request.	Medium
NFR-019	Auditability	The system shall retain request logs for at least 90 days.	Low
NFR-020	Scalability	The system shall meet the performance requirements with 1,000 registered users and 10,000 job records.	Medium
NFR-021	Maintainability	The system shall keep the backend in separate modules (accounts, companies, jobs, job applications, skills) and shall define every API endpoint path in one central configuration file.	Medium
NFR-022	Maintainability	The system shall include at least one automated test for every backend module.	Medium
NFR-023	Compatibility	The system shall work in the two most recent versions of Google Chrome, Microsoft Edge, Mozilla Firefox and Apple Safari.	High
NFR-024	Backup	The system shall create a backup of the main database every 24 hours and keep the last 7 backups.	High
NFR-025	Portability	The system shall run on any host that provides Python 3.13 and PostgreSQL, with the deployment differences set only through environment variables.	Low

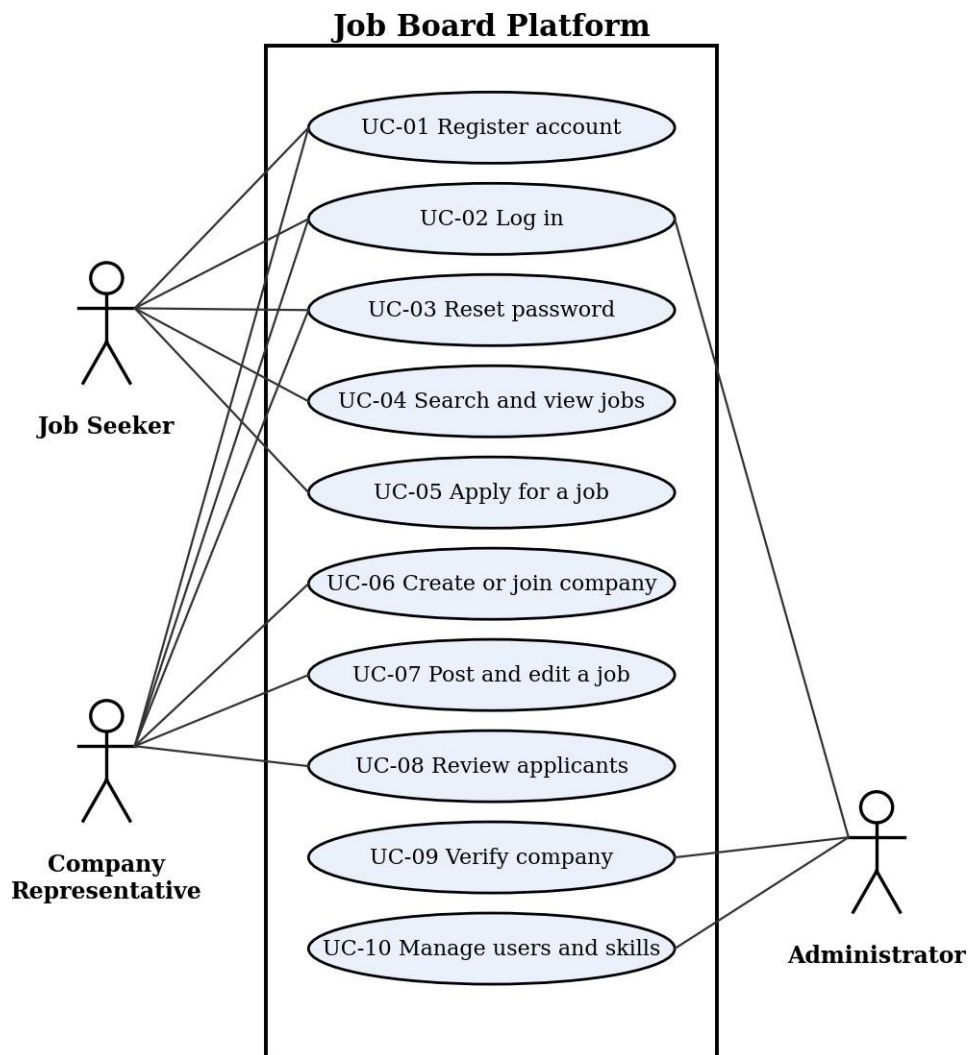
7. Data Requirements and Use Cases

7.1 Data Requirements

Entity	Key data	Validation and storage rules
User	User ID, name, email address, password, role (job_seeker, company_rep, admin), active flag, selected skills.	Email is unique and has a valid format; password is stored as a salted hash and has at least 8 characters; role is one of the three values.
Company	Company ID, name, registration number, verification flag, creation time.	Registration number is required and unique; verification flag starts as false.
Company member	User, company, member role.	Links a Company Representative to a company; used for every ownership check.
Job	Job ID, company, title, description, location, minimum salary, maximum salary, employment type, status, required skills, creation and update time.	Employment type is Full-time, Part-time or Contract; minimum salary is not greater than maximum salary; deletion marks the job inactive.
Job application	Applicant, job, cover letter, resume file, status, submission time.	One application per Job Seeker and job; resume is a PDF of at most 5 MB; status is Pending, Reviewed, Accepted or Rejected.
Skill	Skill ID, name, active flag.	Name is unique ignoring letter case; deletion marks the skill inactive.
Request log	Correlation ID, method, path, status code, duration, request and response body and headers, time.	Stored in the separate log database; passwords and tokens are replaced by a redaction marker; kept at least 90 days.

7.2 Use Case Summary

ID	Use case	Primary actor
UC-01	Register account	Visitor (Job Seeker or Company Representative)
UC-02	Log in	Job Seeker, Company Representative, Administrator
UC-03	Reset password	Job Seeker, Company Representative
UC-04	Search and view jobs	Job Seeker
UC-05	Apply for a job	Job Seeker
UC-06	Create or join a company	Company Representative
UC-07	Post and edit a job	Company Representative
UC-08	Review applicants	Company Representative
UC-09	Verify a company	Administrator
UC-10	Manage users and skills	Administrator



Use case diagram of the Job Board Platform

7.3 Detailed Use Cases

UC-01: Register account

Use case ID	UC-01
Primary actor	Visitor (Job Seeker or Company Representative)
Preconditions	The visitor is not logged in and the email address is not registered.
Main flow	1. The visitor opens the registration page of the chosen role. 2. The visitor enters name, email address and password. 3. The visitor submits the form. 4. The system validates the data and creates the account with the chosen role. 5. The system logs the user in and opens the main page of the role.
Alternative flow	- A1. The email address is already registered: the system shows an error and creates no account. - A2. A field is invalid: the system highlights the field and keeps the form open.
Postcondition	The account exists with the correct role and the user has an active session.

UC-03: Reset password

Use case ID	UC-03
Primary actor	Job Seeker or Company Representative
Preconditions	The account exists and the e-mail service is available.
Main flow	1. The user opens the Forgot password page and enters the email address. 2. The system shows the confirmation message and sends a reset link. 3. The user opens the link and enters a new password. 4. The system stores the new password as a hash and opens the login page.
Alternative flow	A1. The link is invalid or expired: the system shows an error and offers a new request. A2. The email is unknown or belongs to an Administrator: the system shows the same confirmation and sends nothing.
Postcondition	The new password works and the old password no longer works.

UC-05: Apply for a job

Use case ID	UC-05
Primary actor	Job Seeker
Preconditions	The Job Seeker is logged in, the job is open and the Job Seeker has not applied to it before.
Main flow	1. The Job Seeker opens the job list and selects a job. 2. The Job Seeker chooses Apply. 3. The Job Seeker enters a cover letter and attaches a PDF resume. 4. The Job Seeker submits the form. 5. The system validates the data, stores the application with the status Pending and shows a confirmation. 6. The application appears in My Applications.
Alternative flow	A1. The file is not a PDF or is larger than 5 MB: the system shows an error and keeps the form open. A2. The job was applied to before or was closed in the meantime: the system rejects the application and shows a message.
Postcondition	The application is stored and visible to the Company Representatives of the job's company.

UC-07: Post and edit a job

Use case ID	UC-07
Primary actor	Company Representative
Preconditions	The representative is logged in and belongs to a verified company.
Main flow	1. The representative opens the Company dashboard and chooses Post job. 2. The representative enters title, description, location, salary range, employment type and required skills. 3. The representative submits the form. 4. The system validates the data and stores the job with the status Open. 5. The job appears in the job list of the company and in the job list of Job Seekers.
Alternative flow	A1. The company is not verified: the system blocks the action and shows the verification notice. A2. A field is invalid, for example minimum salary above maximum salary: the system highlights the field.
Postcondition	The job is stored and visible to Job Seekers until it is closed or deleted.

UC-08: Review applicants

Use case ID	UC-08
Primary actor	Company Representative
Preconditions	The representative is logged in and at least one application exists for a job of the company.
Main flow	1. The representative opens the Company dashboard and chooses Applicants for a job. 2. The system lists the applicants with name, email address and cover letter. 3. The representative opens the resume of an applicant. 4. The representative selects a new status. 5. The system saves the status.
Alternative flow	A1. The application belongs to another company: the system rejects the request with HTTP 403. A2. The resume file is missing: the system shows a message and the status can still be changed.
Postcondition	The new status is stored and visible to the Job Seeker in My Applications.

UC-09: Verify a company

Use case ID	UC-09
Primary actor	Administrator
Preconditions	The Administrator is logged in and at least one unverified company exists.
Main flow	1. The Administrator opens the Companies page of the admin dashboard. 2. The system lists the companies that await verification with name and registration number. 3. The Administrator reviews a company and chooses Verify. 4. The system sets the verification flag to true. 5. The representatives of the company can now post jobs.
Alternative flow	A1. The Administrator chooses Unverify: posting of jobs is blocked again. A2. A user without the Administrator role sends the request: the system rejects it with HTTP 403.
Postcondition	The verification flag of the company matches the decision and the request is written to the log.

8. Constraints

Category	Constraint
Technology	Backend in Python 3.13 with Django 6.0, Django REST Framework and SimpleJWT; database PostgreSQL; frontend in React with Vite, Tailwind CSS and TanStack Query; CMS in Payload CMS.
Design	Frontend and backend live in separate folders so that they can be deployed separately. The Payload CMS cannot use the business tables directly because its PostgreSQL adapter requires an “id” key that differs from the key names of the Django tables; the CMS therefore works through the backend API.
Hosting	The frontend runs on Vercel and the backend on Railway with limited resources. Railway blocks outbound SMTP ports on the lower plans, so e-mail is sent through an HTTPS-based e-mail API.
Authentication	Google Sign-In works only for origins registered in the Google Cloud console.
Legal and policy	A resume contains personal data and shall be visible only to the applicant and to the company that received the application.
Budget	The project uses free or trial service plans and needs no paid licence.
Schedule	The work follows the timetable of the Software Construction course and of the three-month ByteCorp trainee program.

9. Assumptions and Dependencies

9.1 Assumptions

- Users have a current browser, an internet connection and an e-mail address that they can read.
- Job Seekers have a resume in PDF format.
- Company Representatives know the registration number of their company.
- An Administrator reviews new companies within two working days.
- The deployment team creates the first Administrator account before the first use.

9.2 Dependencies

- Google Identity Services for Google Sign-In.
- An e-mail delivery service for password-reset messages.
- The availability of Vercel, Railway and the PostgreSQL service.
- The open-source libraries named in section 8; a change of their major version may change the code.
- The availability of the backend API for the Payload CMS admin panel.

10. Appendices and Glossary

Appendix A: Traceability Matrix

The matrix connects selected requirements to their source, priority and test case.

ID	Requirement	Source	Priority	Test
FR-001	Register a Job Seeker	Job Seeker	Critical	TC-001
FR-005	Reject login on wrong role page	Administrator	High	TC-004
FR-006	Issue access and refresh tokens	Administrator	Critical	TC-006
FR-010	Send password-reset link	Job Seeker	High	TC-010
FR-026	Block unverified company from posting	Administrator	Critical	TC-014
FR-030	Create a job	Company Rep	Critical	TC-018
FR-041	Filter jobs by employment type	Job Seeker	High	TC-022
FR-046	Apply with cover letter and resume	Job Seeker	Critical	TC-026
FR-047	Accept only PDF up to 5 MB	Company Rep	High	TC-027
FR-054	Set application status	Company Rep	Critical	TC-031
FR-065	Keep CMS and dashboard data identical	Administrator	High	TC-036
NFR-006	Enforce role access on every endpoint	Administrator	Critical	TC-040
NFR-001	Read response within 1 second for 95% of requests	Job Seeker	High	TC-044

Change-management steps

Step	Action
1	Record the change request with its reason and requester.
2	Assess the impact on requirements, design, code and tests.
3	Approve or reject the request.
4	Update the SRS and increase the version number.

5	Notify the team members and the mentor.
6	Revise the affected test cases.

A requirement keeps its identifier when an approved change creates a new version of the SRS, so it stays traceable.

Appendix B: Glossary

Term	Definition
Access token	A short-lived JSON Web Token (15 minutes) that proves the identity of a logged-in user in each API request.
API	Application Programming Interface: the set of endpoints through which the frontend and the CMS use the backend.
CMS	Content Management System; here Payload CMS, used as second admin interface.
Company Representative	A user who acts for a company and posts jobs and reviews applicants for it.
Correlation ID	A unique identifier attached to one API request so that all its log entries can be found.
CORS	Cross-Origin Resource Sharing: browser rule that limits which websites may call the API.
JWT	JSON Web Token: a signed token used for authentication.
Job Seeker	A user who searches for jobs and applies to them.
ORM	Object-Relational Mapper: library that reads and writes database rows as program objects.
RBAC	Role-Based Access Control: access rights that depend on the role of the user.
Refresh token	A long-lived token (7 days) used to obtain a new access token.
REST	Representational State Transfer: an architectural style for web APIs that use HTTP methods and JSON.
Soft delete	Deletion by marking a record inactive instead of removing it from the database.
SPA	Single-page application: a web application that loads once and updates the page in the browser.
SRS	Software Requirements Specification.
Verified company	A company that an Administrator has confirmed and that may therefore publish jobs.